

PRAJWAL BHARADWAJ

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EDUCATION

Doctor of Philosophy in Aerospace Engineering Aug 2026 (Expect.)

Worcester Polytechnic Institute, Worcester, MA

- **Dissertation:** “*Multiscale Mechanics of Tunable Hierarchical Granular Metamaterials for applications in Impact Mitigation*”
- **Advisor:** [Dr. Nikhil Karanjgaokar](#)
- **Research Focus:** Data-driven Material Design and Modeling, Experimental Solid Mechanics, Mechanical Metamaterials, Applied Physics.

Master of Science in Mechanical Engineering

May 2020

Worcester Polytechnic Institute, Worcester, MA

Bachelor of Engineering in Mechanical Engineering

August 2017

Visvesvaraya Technological University, Karnataka, India

RELEVANT EXPERIENCE

Graduate Research Assistant

August 2021 – Current

Aerospace Structures and Materials Lab, Worcester Polytechnic Institute

- **Experimental Instrumentation:** Designed and commissioned multiple bench-top testing setups to characterize specimens under various strain rates (quasi-static to dynamic). This included a gravity-driven drop tower for low-velocity impact and a gas-propelled pneumatic launcher for ballistic-regime characterization, both fully integrated with high-speed optical diagnostics, integrated load cells and data acquisition hardware for high-fidelity mechanical testing.
- **Optical Metrology:** Developed custom non-contact optical measurement techniques utilizing high-speed imaging, **hierarchical segmentation** and **Digital Image Correlation (DIC)** to extract full-field displacement data from highly deformable discrete metamaterials undergoing high-strain rate loading.
- **Computational Modeling (FEM):** Developed a comprehensive suite of simulation tools in **Abaqus Python** to characterize metamaterials across multiple length scales (micro-contact to macro-chain) and strain rates. This framework enables rapid virtual testing of both continuous and discrete metamaterials, bridging the gap between discrete particle dynamics and continuum material response.
- **Data Science & Machine Learning:** Developed semi-automated data pipelines for the extraction, processing, and visualization of large experimental datasets; utilized design of experiments (DoEx) and statistical learning to build reduced-order models (ROMs), successfully identifying key performance drivers in high-dimensional parameter spaces.

R&D Mechanical Engineer

Jan.2020 – Dec. 2020

Multiscale Systems, Worcester, MA

- Contributed to confidential R&D projects focused on design, fabrication and testing of materials for applications in geothermal engineering; specific project details and outcomes are proprietary/embargoed due to funding mandates.
- Collaborated with a cross-functional team to advance technology readiness levels for next-generation geothermal energy systems.

TECHNICAL SKILLS

Experimental Methods: Universal Testing Systems (Instron), Gravity-Driven Impact Testing, Pneumatic Impact Testing, Split-Hopkinson Pressure Bar, Laser Vibrometry and High-Speed Imaging

DS & ML Libraries: NumPy, SciPy, Pandas, Scikit-learn, Pytorch, Matplotlib, Seaborn, Open-CV

Analysis Software: Abaqus/Standard, Abaqus/Explicit, ANSYS Mechanical

Design Software: SolidWorks, OnShape, Fusion360, SpaceClaim DM, nTopology

Programming Languages: Python, MATLAB

Additive Manufacturing: Stereolithography (SLA), Fused Deposition AM (FDM), Laser Powder Bed Fusion (L-PBF)

Computational Methods: Finite Element Method (FEM), Finite Difference Method (FDM), Discrete Element Method (DEM)

Others : L^AT_EX, MS Office Suite, InkScape

JOURNAL ARTICLES

In Press

- [1] Kodali, M., **Bharadwaj, P.**, Bhavanam, S., & Karanjgaokar, N. (2023). Semi-analytical approach to determine interfacial adhesion in polymer-bonded explosives. *Experimental Mechanics*, 63(2), 237-250.
- [2] Olleak, A., Dugast, F., **Bharadwaj, P.**, Strayer, S., Hinnebusch, S., Narra, S., & To, A. C. (2022). Enabling Part-Scale Scanwise process simulation for predicting melt pool variation in LPBF by combining GPU-based Matrix-free FEM and adaptive Remeshing. *Additive Manufacturing Letters*, 3, 100051.

In Review

- [1] **Bharadwaj, P.**, Kocharyan, H., & Karanjgaokar, N. Dynamic Behavior of 2D Hierarchical Granular Metamaterial Fabrics. (*Submitted*, June. 2026).
- [2] **Bharadwaj, P.**, Saidy, K., Kocharyan, H., Yeghiazaryan, V., Yuan, Z., & Karanjgaokar, N. Architected for Attenuation: Hierarchical Granular Metamaterial Chains for Impact Mitigation. (*Submitted*, May. 2026).

In Preparation

- [1] **Bharadwaj, P.**, Saidy, K., Kocharyan, H., Yuan, Z., & Karanjgaokar, N. Towards the Design of Magnetoactive Re-Entrant Auxetic Metamaterials for Adaptive Energy Absorption. (*Target Submission: August 2026*).

CONFERENCE PRESENTATIONS

- [1] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Conference: 2026 SEM Annual Conference & Exposition on Experimental & Applied Mechanics, Track: Multiphysics of Metamaterials, held at Norfolk, Virginia, June 1-4 2026.
- [2] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Conference: 20th European Mechanics of Materials Conference, Track: Mechanics of Architected Materials: Experiments and Modeling, held at Florence, Italy, May 27-29 2026.
- [3] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Conference: 2025 SES Annual Technical Meeting, Track: Controlling Mechanical Waves with Metamaterials, held at Atlanta, Georgia, October 12-15 2025.
- [4] **P. Bharadwaj**, K. Saidy, H. Kocharyan, Z. Yuan, N. Karanjgaokar, Conference: 12th European Solid Mechanics Conference, Track: Architected and Additively Manufactured Materials, held at Lyon, France, July 7-11 2025.
- [5] **P. Bharadwaj**, K. Saidy, H. Kocharyan, Z. Yuan, N. Karanjgaokar, Conference: 2025 SEM Annual Conference & Exposition on Experimental & Applied Mechanics, Track: Additive and Advanced Manufacturing, held at Milwaukee, Wisconsin, June 2-5 2025.
- [6] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Conference: 2023 SEM Annual Conference & Exposition on Experimental & Applied Mechanics, Track: Additive and Advanced Manufacturing, held at Orlando, Florida, June 5-8 2023.
- [7] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Conference: 2025 SES Annual Technical Meeting, Track: Controlling Mechanical Waves with Metamaterials, held at College Station, Texas, October 16-21 2022.
- [8] **P. Bharadwaj**, N. Karanjgaokar, Conference: 2022 SEM Annual Conference & Exposition on Experimental & Applied Mechanics, 23rd International Symposium on Micro- and Nanomechanics (ISMAN) held at Pittsburgh PA, 13-16 June 2022.
- [9] **P. Bharadwaj**, N. Karanjgaokar, Conference: 2022 SEM Annual Conference on Experimental and Applied Mechanics, Additive and Advanced Manufacturing held at Pittsburgh PA, 13-16 June 2022.
- [10] **P. Bharadwaj**, N. Karanjgaokar, Conference: 2022 SEM Annual Conference on Experimental and Applied Mechanics, Dynamic Behavior of Materials to be held at Pittsburgh PA, 13-16 June 2022.

POSTER PRESENTATIONS

- [1] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Event: 2026 Northeastern SEM Student Symposium, to be held at UMass, Dartmouth, Massachusetts, April 11th 2026.
- [2] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Event: 2026 Graduate Research Innovation Exchange, to be held at WPI, Worcester, Massachusetts, February 20th 2026.
- [3] **P. Bharadwaj**, H. Kocharyan, N. Karanjgaokar, Event: 2025 Graduate Research Innovation Exchange, held at WPI, Worcester, Massachusetts, February 21st 2025.
- [4] **P. Bharadwaj**, N. Karanjgaokar, Event: 2023 Graduate Research Innovation Exchange, held at WPI, Worcester, Massachusetts, February 21st 2023.

TEACHING & MENTORSHIP

Teaching

Graduate Teaching Assistant

2025 – Present

Worcester Polytechnic Institute, Aerospace Engineering Department

Flagship Assignment: AE3010 Engineering Experimentation and Data Science

- [1] **Curriculum Innovation:** Conceptualized and implemented a *pre*, *-post* and a *final measurement design project* as formative and summative assessments, requiring students to design independent experimental protocols and validate sensor data.
- [2] **Technical Instruction:** Led tutorials on **LabVIEW** and **MATLAB** for signal processing and data analysis, directly supporting the course's data science initiative.
- [3] **Mentorship:** Managed a cohort of **30** students, providing weekly technical guidance on instrumentation, measurement and analysis.

Additional Teaching Assignments:

- [1] **AE2310: Introduction to Aerospace Control Systems:** Led recitation sessions on dynamics and controls and graded technical reports for 51 students.
- [2] **AE3210: Fundamental of Aerodynamics:** Led recitation sessions on introductory aerodynamics and graded technical reports for 90 students.
- [3] **AE4210: Fundamentals of Air-Breathing Propulsion:** Led recitation sessions on dynamics and controls and graded technical reports for 45 students.

Research Mentorship

Graduate Research Mentor (M.S. Level)

2024 – Present

Assisted advisor in the technical supervision of 3 Master's students

- **Technical Training:** Taught students the fundamentals of **mechanical testing, specimen fabrication and finite element modeling**. Reviewed their Python/MATLAB code for efficiency and accuracy.
- **Outcomes:** Students successfully completed multiple semester long directed research/independent research studies, with their results being presented at various conferences.

Undergraduate Research Mentor (SURF Program)

Summer's 22, 23, 2024

Direct supervisor for 2 fellows in the Summer Undergraduate Research Fellowship

- **Project Management:** Defined the scope for 10-week intensive research projects on mechanical metamaterials ensuring achievable milestones within a tight time-frame.
- **Technical Training:** Taught students the fundamentals of **mechanical testing, specimen fabrication and finite element modeling**. Reviewed their Python/MATLAB code for efficiency and accuracy. Moved students from novices to independent operators during the duration of their program.
- **Outcomes:** Students created research presentations presented at the WPI Undergraduate Research Symposium.

AWARDS & NOMINATIONS

- [1] **Finalist, Michael Sutton International Student Paper Competition**, Society of Experimental Mechanics, June 2026.
Awarded \$1,000 in conference funding support to present doctoral research at SEM 2026 Annual Conference and Exposition.
- [2] **Finalist, TA of the Year**, Morgan Teaching and Learning Center, April 2026.
Selected from a highly competitive pool of university instructors for outstanding contributions to undergraduate engineering instruction and mentorship.
- [3] **Alex F. Backlin Fund Scholarship**, Worcester Polytechnic Institute, August 2025.
Awarded \$5,000 in tuition and fee assistance to support the final phase of doctoral research.
- [4] **SEM Student Ambassadorship Award**, Society of Experimental Mechanics, June 2025.
Awarded \$500 for conference registration and recognized for both technical and leadership excellence in experimental mechanics at the 2025 Annual Conference.
- [5] **Graduate Student Travel Award**, National Science Foundation & WPI Aug. & Oct. 2022
Awarded \$1,500 to present research findings at the Society of Engineering Sciences (SES) Annual Technical Meeting in College Station, TX.
- [6] **Tau Beta Pi (National Engineering Honor Society)**, MA Alpha Chapter Nov. 2024
Inducted for distinguished scholarship, leadership, and exemplary character in engineering.
- [7] **Sigma Gamma Tau (Aerospace Engineering Honor Society)**, Northeastern Chapter May 2020
Recognized for outstanding academic achievement and excellence in the field of aerospace engineering.

PROFESSIONAL AFFILIATIONS & SERVICE

- [1] **President & Vice-President**, Graduate Student Government (GSG), WPI Aug. 2021 – May 2023
Led the executive board and senate (40+ members) to advocate for student interests. Spearheaded orientation programs, managed communication channels, and represented the graduate body at university-level town halls.
- [2] **Member**, Society for Experimental Mechanics (SEM) 2021 – Present

REFERENCES

Dr. Nikhil Karanjgaokar

Associate Professor, Aerospace Engineering Department
Worcester Polytechnic Institute

Relationship: Doctoral Advisor

nkaranjgaokar@wpi.edu

Dr. Zhangxian Yuan

Assistant Professor, Aerospace Engineering Department
Worcester Polytechnic Institute

Relationship: Doctoral Committee Member/Collaborator

zyun2@wpi.edu

Dr. Zachary Taillefer

Assistant Professor, Aerospace Engineering Department

Worcester Polytechnic Institute

Relationship: Primary Course Instructor (Experimentation & Data Science)

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